

Name: LeBroners

Date: 04/11/24

Class: 3

MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The solution has to be able to grip tightly onto any drawstring without slipping or falling off

- Currently, there have been no solutions
- The current hoodie designs have holes that are too large for the string, leading to the string getting lost into the hood

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #2 (HIGH Priority)

The solution has to be sturdy and not cause damage to anything it comes in contact with.

- Certain types of materials that go into the dryer get stuck in the small cracks and cause scratches while in the cycle

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #3 (HIGH Priority)

Must make sure that the hoodie string does not get lost.

- This is the main concern right now
- There are no reliable solutions currently to solve the problem

Stakeholder feedback: does not happen that often but will be a cool solution

Name: LeBroners

Date: 04/11/24

Class: 3

Design Requirement #4 (HIGH Priority)

Must look fashionable or cute to wear outside

- Knotting the strings, or attaching the strings does not look nice
- It looks weird to attach safety pins to the hoodie strings

Stakeholder feedback: knot looks okay, does not need to be that high to a priority

Design Requirement #5 (HIGH Priority)

Must be something that can stay on effortlessly

- Remembering to tie the strings together or attach the strings together is too much to remember and people often forget

Stakeholder feedback: Make sure that the item does not cause damage to its surroundings

Design Requirement #6 (LOW Priority)

Be kid-proof and pet-proof

- It cannot be dangerous to kids or pets

Design Requirement #7 (LOW Priority)

Not damage the hoodie string

- If possible, it cannot have any sharp parts that might pierce the string
- Cannot be a pin that will puncture the string

FUTURE TEST REQUIREMENTS

Requirements that we will test for:

1. Size large enough to not get lost in the hoodie
 - Measure the diameter of the hoodie string hole + calculate the circumference
 - Measure+calculate circumference of the prototype

Name: LeBroners

Date: 04/11/24

Class: 3

- Diameter of hoodie string hole < circumference of prototype
- 2. Attachability - grip strength
 - Measure the circumference of the string
 - Measure the circumference of the clip
 - String circumference < circumference clip
- 3. Not cause damage - scratch
 - Scratch device against several surfaces Ex: paper, wood, metal, fabric, clothing
 - If passes all, then successful

Expert Feedback: if we use a plastic or metal, won't it bang against the machine and cause noise

- If plastic, it might melt in the high temperatures of the dryer

Rubric

Element C: Presentation and Justification of Solution Design Requirements						
C1. Determination of design requirements. _____ design requirements are related to prior research.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	No
C2. Requirements list. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	are listed, prioritized, clear, and detailed	are listed, prioritized, generally clear, and detailed	are listed, prioritized, generally clear, and somewhat detailed	are listed and prioritized but some or all may be incomplete	are listed, formatted, and prioritization is attempted,	are not present or are too vague

Name: LeBroners

Date: 04/11/24

Class: 3

				and/or lack specificity	but these may be partial or overly general	
C3. Quantity of high-priority design requirements. _____ of the design requirements listed are identified as high priority.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	Five	Four	Three	Two	One	None
C4. Objective & measurable. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	are consistently objective and measurable	are nearly always objective and measurable	are generally objective and measurable	may be only sometimes objective and/or measurable	are insufficiently measurable	are not measurable
C5. Leading to solution. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	would be highly likely to lead to a tangible and viable problem solution	would be likely to lead to a tangible and viable problem solution	have a potential to lead to a tangible and viable problem solution	have some potential to lead to a tangible and viable problem solution	have little potential to lead to a tangible and viable problem solution	cannot lead to a tangible and viable problem solution
C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _____ primary stakeholder groups.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	no primary stakeholder groups.